

The World Doesn't Run on Code Alone

Assignment

Thursday, May 07, 2026

Topic Proposal & Research Plan (D0) - [Group]	May 20, 2026 @ 1200Hrs
Fieldwork Insights Report (D1) - [Individual]	June 10, 2026 @ 2355Hrs
Data Collection Design & Pilot Report (D2) - [Group]	July 01, 2026 @ 2355Hrs
Data Summary & Preliminary Findings (D3) - [Group]	July 15, 2026 @ 2355Hrs
Final Research Report (D4) - [Individual]	August 03, 2026 @ 2355Hrs
Individual Viva & Presentation (D5) - [Individual]	August 17 - 26, 2026 @ 2355Hrs
Grade Value	40% out of 100% of the module

1 Introduction

Many computing solutions fail not because of poor code, but because they ignore how people behave in real environments. This project challenges you to move beyond assumptions and engage directly with the world your systems are meant to serve. You will observe, question, and measure real situations, working with data that is often incomplete, inconsistent, and messy. The goal is to develop the ability to think beyond code and ground your decisions in real evidence.

2 General Guidelines

The following guidelines are to be adhered to when undertaking this assignment.

2.1 Area of the Field Study

This assignment requires students to investigate real world behaviour and everyday systems within and around the University of Colombo and its surrounding urban environment. The selected research areas must allow direct observation of people, spaces, and activities, and must support both qualitative and quantitative data collection through field engagement such as observation, interviews, counting, or simple measurements. Topics should be

grounded in visible, accessible, and measurable phenomena rather than abstract technological concepts or purely theoretical discussions. The focus is on understanding how people behave, interact, and make decisions in real environments, and how these patterns can be studied systematically using research methods introduced in the course.

Acceptable research areas are as follows:

- Queueing and waiting behaviour in public and institutional spaces
- Human movement and navigation within campus and urban environments
- Space usage patterns in libraries, study areas, and common facilities
- Decision-making under constraints (time, cost, availability)
- Informal systems, workarounds, and rule-bending in everyday processes
- Time use and productivity patterns among students and staff
- Social interaction and group behaviour in shared environments
- Perception of safety and adaptation of behaviour in public spaces
- Transport related behaviour (bus stops, walking routes, commuting patterns)
- Service interaction behaviour in administrative or commercial settings (e.g., canteens, offices, shops)

2.1.1 D0 : Topic Proposal & Research Plan (Group Submission)

This submission defines the foundation of your research project. Each group must clearly identify a real-world problem within the approved domains and present a focused research topic supported by a brief explanation of its relevance. The proposal should include preliminary research questions, the selected field site(s), and a brief justification of why the problem is worth investigating. In addition, groups must outline their intended approach for both qualitative and quantitative data collection, including possible methods such as observation, interviews, surveys, or simple measurements. This submission is intended to ensure that the research is feasible, grounded in reality, and suitable for mixed-method investigation.

2.1.2 D1 : Fieldwork Insights Report (Individual Submission)

This individual report captures each student's direct engagement with the field during the qualitative phase of the study. Students are expected to present structured observations, interview summaries, and early interpretations based on real-world interactions and experiences. The focus is on describing what was observed and heard, identifying emerging patterns in behaviour, and reflecting on initial insights without introducing statistical analysis at this stage. This submission ensures that every student actively participates in fieldwork and develops the ability to interpret real-world phenomena using qualitative research techniques.

2.1.3 D2 : Data Collection Design & Pilot Report (Group Submission)

In this submission, groups are required to design and validate their quantitative data collection instruments based on insights gained from the qualitative phase. This includes developing surveys, defining variables, and ensuring that questions are clear, measurable, and aligned with the research objectives. Groups must also conduct a pilot test with a small sample to identify issues such as ambiguity, bias, or ineffective question design. A reflection on the pilot process and necessary revisions must be included. This step ensures that the quantitative phase is methodologically sound and grounded in prior qualitative findings.

2.1.4 D3 : Data Summary & Preliminary Findings (Group Submission)

This submission presents the collected quantitative data in a structured and meaningful way. Groups are expected to summarise their dataset using appropriate descriptive statistics such as frequencies, percentages, and basic comparisons where relevant. The focus is on identifying early patterns and trends in the data rather than performing advanced statistical analysis. Students should also begin linking these findings back to the qualitative insights collected earlier, highlighting confirmations, contradictions, or new emerging patterns.

2.1.5 D4 : Final Research Report (Individual Submission)

The final report is a comprehensive individual submission that integrates all phases of the research process. Students must combine qualitative insights, quantitative findings, and literature-based understanding into a coherent analysis of the selected problem. The report should clearly demonstrate how field observations informed data collection, how quantitative results supported or challenged initial findings, and what overall conclusions can be drawn. Students are also expected to discuss the implications of their findings for real-world systems and reflect on the strengths and limitations of their research approach.

2.2 Grouping Criteria

The students will work in groups of 3 to 4 members. The formation of the group should be based on a shared research interest within the approved topic domains to ensure coherence in fieldwork and data collection. Each group must select a research focus that allows all members to participate actively in observation, interviews, and survey-based data collection. Groups are expected to distribute field responsibilities evenly, but each student must also demonstrate individual contribution through personal field notes and individual submissions where required. Group composition will be fixed after approval of the topic proposal and cannot be changed thereafter.

2.3 Submission

Students should undertake and conduct one of the aforementioned field studies and submit the required documents by the deadlines specified above. The submissions should be made to the LMS on or before the specified deadline.

2.4 Plagiarism

All submissions will be checked for plagiarism using relevant software. The students must have a plagiarism ratio of less than 5% without the references section. Any submission

having a plagiarism score above 5% is considered to have substantial plagiarism and would be awarded penalties. However, a plagiarism ratio above 15% is regarded as highly plagiarized work and **awarded zero (0)** marks.

2.4.1 AI Based Report Generation

If a student engages in any form of report generation through AI tools (e.g. ChatGPT) in whole or part, the student will fail the course irrespective of the final examination. A 'D' grade will be awarded to such a student.

3 Documentation Guidelines

- This assignment requires students to use Harvard notation to cite work referred to in the submission. Therefore, using any other referencing styles, such as IEEE, APA, or Oxford, is unacceptable for the purpose of this assignment.
- This assignment requires students to use LATEX as the primary tool in compiling the written submissions. **Submissions compiled through any other word-processing system will not be accepted for the purpose of this assignment.**
- The report submissions should have a maximum of 2000 words, excluding captions, tables, footnotes, acknowledgement, abstract, table of content, lists of figures, tables and acronyms, and references. No minimum word count will be enforced, and it is entirely up to the individual in the case of individual submissions and the group in the case of group submissions to decide on the length of each submission. However, not having a large number of words is not directly proportional to the quality of the work and, thus, the grade.
- The following structure needs to have adhered to when compiling the report:
 - Title Page
 - Acknowledgement
 - Abstract
 - Table of Contents
 - List of Figures
 - List of Tables
 - List of Acronyms
 - Introduction Chapter
 - Any number of Chapters the student or the group deems relevant
 - Critical Analysis
 - Conclusion and Future Work
 - List of References
 - Index
- The submissions that do not follow the format above will be penalized accordingly.

- The students can use any fonts of their choice. However, the font size should be 12pt . In addition, the dimensions of the headings and titles should be adjusted accordingly.
- Page margins should be Left: 1.5", Right: 1", Top: 1.5", Bottom: 1"
- Page numbering should appear in the centre of the footer of each page without the title page . The Roman numbers (e.g. i, ii, iii) will be used from the first page after the title page to the last page before the introduction chapter. The Arabic numerals (e.g. 1, 2, 3) are to be used from the first page to the introduction chapter until the end of the document.
- There should be no numbering of sections until Chapter One of the document. The list of references and Index section should not have section numbering.
- The “Critical Analysis” Chapter should critically review the area studied. This should necessarily include the students’ formulations, opinions and criticisms.
- The “Conclusions and Future Work” Chapter should summarise the survey and point out areas of future research that could advance the areas.
- The submissions should have bookmarks for easy navigation.
